Project Title: ESP32-Based Intelligent Multi-Port Battery Management System

1. Brief Description of the Project:

1.1 Project Background

Rechargeable batteries are widely used in mobile devices, electronics, renewable energy systems, and electric vehicles. However, basic battery chargers usually provide limited monitoring and control, and they may not show the battery's real-time voltage, current, and temperature.

This project proposes a smart 3-port battery charging and management system using ESP32. Each port will independently monitor the battery's voltage, current, and temperature and control charging through a relay. The system will provide both Manual Mode and Automatic Mode, allowing the user to control charging directly or let the system automatically manage it based on predefined safety limits.

The collected battery data will also be analyzed using Python and Machine Learning to monitor charging behaviour, detect abnormal conditions, and evaluate changes in battery condition over repeated charging cycles. Thus, the project combines battery charging, real-time monitoring, automation, IoT, and Machine Learning in a single low-cost system.

1.2 Objectives and Aims of the Project

1. Design and develop an independent 3-port battery charging system with voltage, current, and temperature sensing.
2. Develop ESP32-based control for real-time monitoring and relay operation in both Manual and Automatic modes.
3. Develop a web-based dashboard for live monitoring of battery voltage, current, temperature, charging status, and other parameters.
4. Implement a Machine Learning system using Python and Scikit-learn to analyze battery data, detect abnormal charging behaviour, and monitor battery condition over multiple charging cycles.

2. Program Outcomes (PO) Mapping

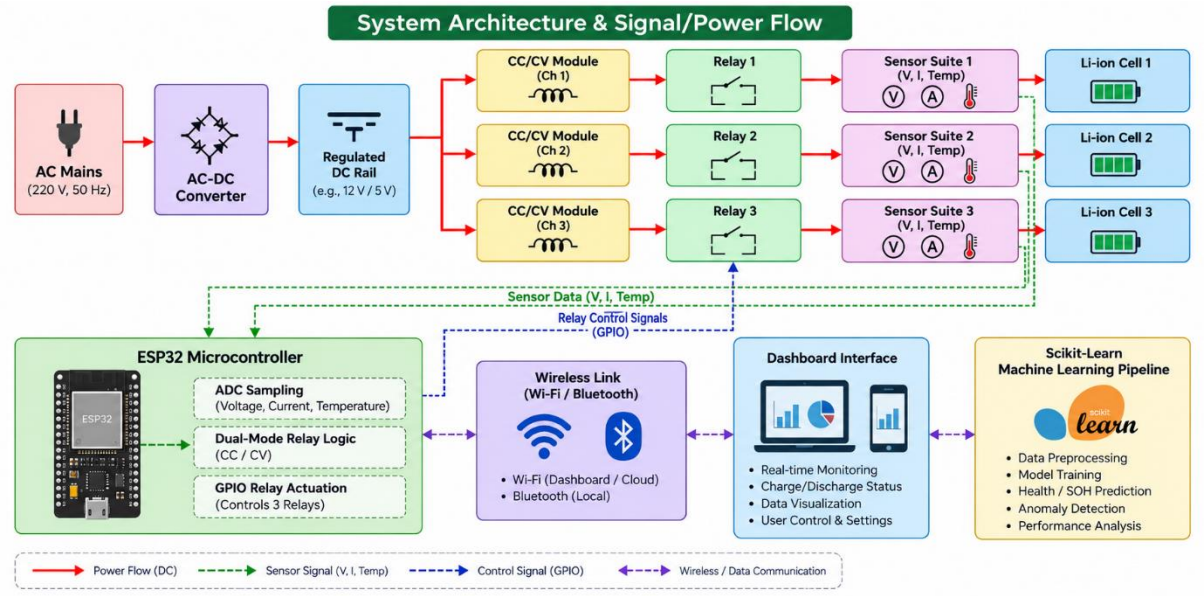
Program Outcome (PO)	Description & Project Mapping
PO(e): Modern Tool Usage	The project uses modern tools such as ESP32 and Arduino IDE for hardware control, Python, NumPy, Pandas, and Scikit-learn for data analysis and Machine Learning, and Flask/Streamlit for dashboard development and data visualization.
PO(i): Individual Work & Teamwork	Team members will share responsibilities in different areas, including hardware development, ESP32 programming, dashboard development, Machine Learning, and system testing. The project will require both individual contributions and teamwork for successful integration.
PO(j): Communication	The project will involve technical discussions, progress presentations, report writing, and final project demonstrations. Team

	members will communicate their work and results clearly through presentations, documentation, and a project poster.
PO(k): Project Management & Finance	The project will be completed according to a planned timeline with different development milestones. Component selection, budgeting, purchasing, testing, and prototype development will be managed within the estimated budget of 3,800–6,700 BDT.

3. Project Methodology

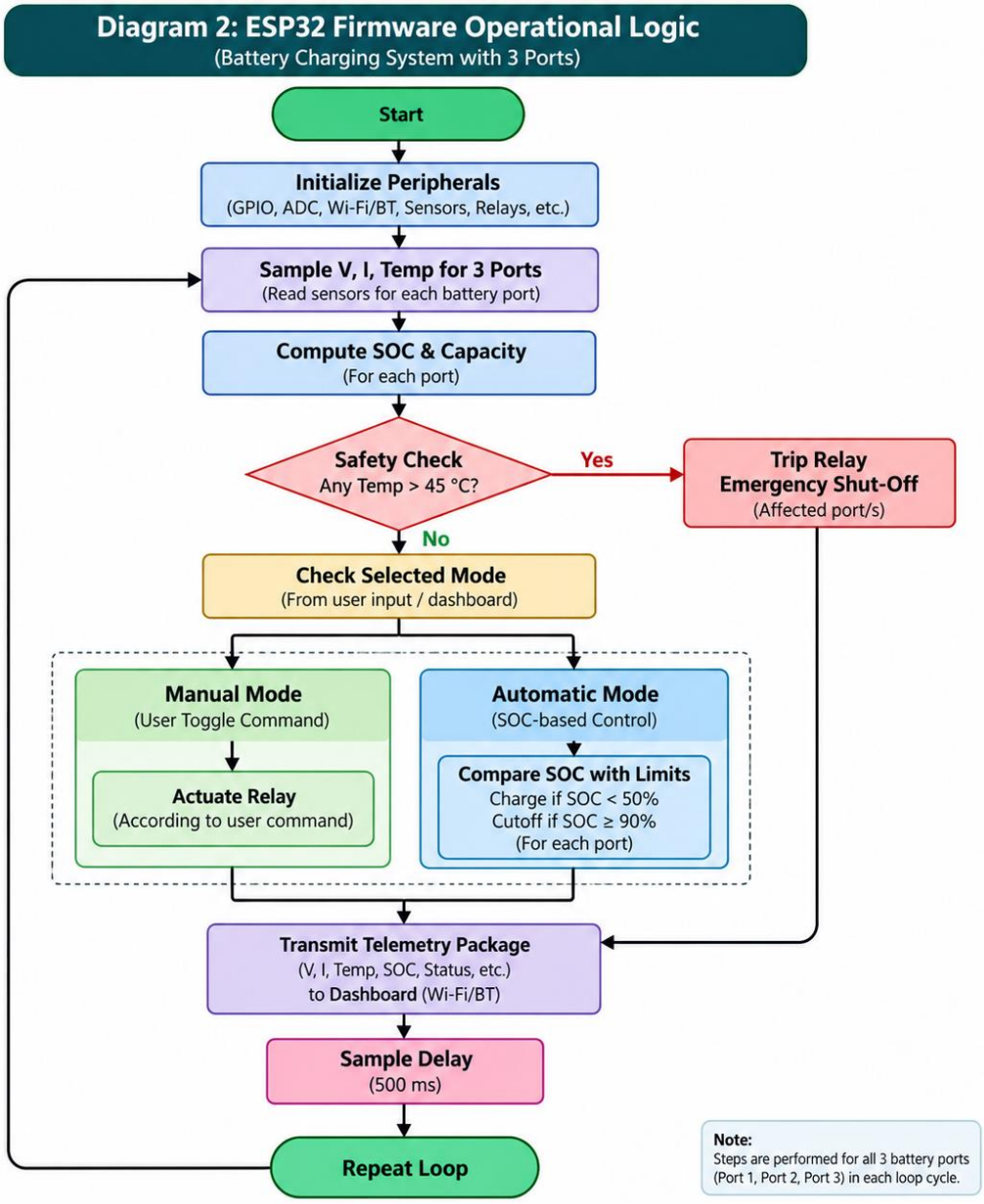
3.1 Hardware Architecture & Sensing:

1. **Power & Charging Stage:** AC mains input converted to regulated DC rails feeding independent CC/CV charging modules set to a 4.2 V termination limit per Li-ion channel.
2. **Relay Switching & Sensing:** Each of the 3 channels includes an independent relay for isolation, along with voltage divider circuits, current sensors (ACS712 / INA219), and temperature sensors (DS18B20 / NTC thermistor) connected to the ESP32 ADC/digital pins.



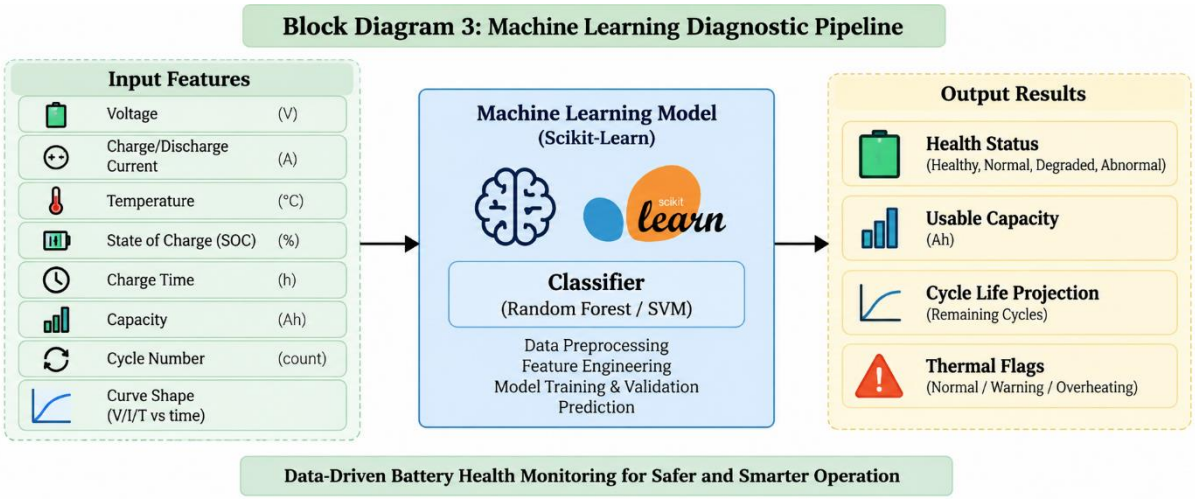
3.2 Firmware & Control Logic (ESP32):

- **Sensing Loop:** Continuous acquisition of battery voltage, charge/discharge current, temperature, charging time, discharging time, and cycle count.
- **Dual Operating Modes:**
  - *Manual Mode:* Direct ON/OFF toggle of individual port relays via dashboard commands.
  - *Automatic Mode:* ESP32 automatically energizes/de-energizes relays based on user-configured thresholds (e.g., start charging when SOC < 50%, stop when SOC ≥ 90%, or trip on temperature safety limit > 45 °C).



3.3 Software Dashboard & ML Pipeline:

- **Telemetry Transmission:** Real-time data streamed from ESP32 to smartphone/laptop over Bluetooth/Wi-Fi via WebSockets.
- **Machine Learning Pipeline (Python):** Sensor data features (voltage, current, temperature, SOC estimate, capacity delivered, cycle number, and curve shape) are processed using algorithms like Random Forest, SVM, or ANN to classify battery status into **Healthy**, **Normal**, **Degraded**, or **Abnormal**.
- **Dashboard Interface:** Renders live voltage/current gauge displays, multi-port SOC history graphs, ML condition flags, and threshold controls.



4. Expected Results

- **Safe Independent Charging:** Precise CC/CV charging across all 3 ports with automatic relay disconnect upon reaching full charge or safety limits.
- **Real-Time Telemetry & Dual-Mode Control:** Seamless wireless monitoring and low-latency automated relay switching based on user-defined SOC thresholds.
- **Accurate Health Assessment:** Machine learning classification of battery health status with clear graphical degradation trend tracking across charge-discharge cycles.
- **User Alerting:** Visual dashboard flags and system warnings for degraded or abnormally behaving batteries.

5. Relevance to National Development

Rechargeable batteries are widely used in Bangladesh for easy bikes, solar systems, mobile devices, and other electronic equipment**.** Poor charging and battery management can reduce battery life, increase replacement costs, and create safety risks. This project aims to develop a **low-cost,** locally buildable smart charging and monitoring system that improves battery safety, reduces unnecessary battery replacement, and supports efficient energy use.

6. Basic Facilities Available in the Department

- **Microprocessor & Embedded Systems Laboratory:** ESP32 programming, testing, and debugging facilities.
- **Electronics & Power Electronics Laboratory:** DC power supplies, oscilloscopes, mustimeters, soldering stations, breadboards, and PCB prototyping facilities.

7. Key Performance Indicators (KPIs)

- **Voltage & Current Accuracy:** Reliable measurement of battery voltage and charging current.
- **Temperature Monitoring:** Accurate detection of abnormal temperature conditions.
- **Automatic Control:** Fast relay response when safety limits are exceeded.

- **Charging Protection:** Automatic charging cutoff during unsafe conditions.
- **ML Performance:** Reliable classification of battery condition.
- **Wireless Monitoring:** Fast and reliable data updates on the dashboard.

8. Research Scope

- Machine Learning-based **SOC and SOH estimation** of lithium-ion batteries is an active research area.
- Recent research uses **voltage, current, temperature, and charging-cycle data** for battery condition estimation.
- Related paper: *"State-of-Health and State-of-Charge Estimation in Electric Vehicles Batteries: A Survey on Machine Learning Approaches"* — IEEE Access, 2024.
- **Research link:** IEEE Xplore Paper
- This project applies these research concepts to a **low-cost 3-port smart charging and monitoring system**.

9. Cost Estimation

Sl. No.	Name of Item / Description	Quantity	Unit Price (BDT)	Total Cost (BDT)
1	ESP32 Microcontroller Board	1	750	750
2	Independent CC/CV Charging Modules (e.g., TP4056 / Buck CC-CV)	3	120	360
3	3-Channel Relay Module (5V/10A)	1	250	250
4	Current Sensors (ACS712 / INA219)	3	220	660
5	Voltage Divider / Voltage Sensing Circuits	3	50	150
6	Temperature Sensors (DS18B20 / NTC)	3	120	360
7	AC-to-DC Power Supply Adapter (12V/5V DC)	1	650	650
8	3.7 V Li-ion Test Batteries & Battery Holders	3	300	900
9	Enclosure, Connectors, PCB, and Wiring	1 Lot	500	500
Total	**Estimated Prototype Cost**			**4,580 BDT**

Signatures :

Project Members

Sl. No.	Name	Roll No.	Signature	Date
1	**Md. Toufikur Rahman**	**2201014**	_____	_____
2	**Md. Najmus Shakib**	**2201151**	_____	_____

Supervisor

Name & Designation	Signature	Date
Dr. G. K. M. Hasanuzzaman Professor, Department of EEE, RUET	_____	_____